Workshop – Week 4 – Some selected solutions

1. If a random number generator is producing numbers in the interval [55, 62) can it ever produce the number 62? Explain your answer.

Solution:
NO, the brackets indicate that the range starts with 55 and ends just before 62. If the numbers are integer values, then the largest possible number in the sequence is 61.

3. Write a program that generates random numbers in the interval [2, 12]. Generate 6000 numbers and display the results for every number that can occur in the interval.

Solution:

```
#include <iostream>
#include <cstdlib>
using namespace std;

int RandomInt(int first, int last);

int total;
int count[15];

//-----------------------------------------------------------------
int main() {
  int roll, i;

  srand(683);  // the seed - what happens if you change it?

  for (i = 0; i < 15; i++) {
    count[0] = 0;
  }

  total = 6000;
  cout << "Generating " << total << " random numbers from 2 to 12\n";
  for (i = 0; i < total; i++) {
    roll = RandomInt(2, 12);
    if ((roll >= 2) && (roll <= 12)) {
      count[roll]++;
    } else {
      cout << "Illegal roll of " << roll << endl;
    }
  }

  cout << "\nResults:\n";
  for (i = 2; i <= 12; i++) {
    cout << "Rolled " << i << ": " << count[i] << " times.\n";
  }
  cout << endl;
}

//-----------------------------------------------------------------
int RandomInt(int first, int last) {
// returns a random number in the range [first, last]
```

```
  int diff;
  diff = (last - first) + 1;
  return rand() % diff + first;
}
```

7. Write the correct MASSEY machine code instructions to carry out the following:
a) place the hexadecimal value 23 in register 0 and also in a memory location
b) OR the contents of registers 8 and 9 and store the result in a memory location
c) add the contents of two memory locations
d) jump to the instruction at memory address 13 if register 5 contains a 6

Solutions:

a)
```
1023
4036
```
Note: you must select a memory location – I selected location 36.

b)
```
B189
4145
```
Note: I selected to use R1 for the result and I selected memory location 45.

9. Divide the instructions below into two groups as follows:
Group A – instructions that involve memory location 2A
Group B – instructions that have nothing to do with memory location 2A

112A 202A 332A 442A 552A 662A 882A DD2A

Solution:

Group A: 332A 442A DD2A

Group B: 112A 202A 552A 662A 882A